

MANUFACTURING INSTRUCTION (NATURAL LANGUAGE)

Pick the blue box from shelf A, place it on workstation 2, verify placement, and retry if the object is missing.

Generate workflow

Pick and place: blue box from shelf A to workstation 2

"Pick the blue box from shelf A, place it on workstation 2, verify placement, and retry if the object is missing."

PARSED VIA DETERMINISTIC FALLBACK

STRUCTURALLY VALID

Fallback reason: Gemini API call failed: 429 You exceeded your current quota, please check your plan and billing details. For more information on this error, head to: <https://ai.google.dev/gemini-api/docs/rate-limits>. To monitor your current usage, head to: <https://ai.dev/rate-limit>. * Quota exceeded for metric: generativelanguage.googleapis.com/generate_content_free_tier_requests, limit: 5, model: gemini-3.0-flash Please retry in 55.545138424s. [links { description: "Learn more about Gemini API quotas" url: "https://ai.google.dev/gemini-api/docs/rate-limits" } , violations { quota_metric: "generativelanguage.googleapis.com/generate_content_free_tier_requests" quota_id: "GenerateRequestsPerMinutePerProjectPerModel-FreeTier" quota_dimensions { key: "model" value: "gemini-3.0-flash" } quota_dimensions { key: "location" value: "global" } quota_value: 5 } , retry_delay { seconds: 55 }]

ACTION: Locate blue box at shelf A

tools: vision_system

failure modes: **object_missing**

↳ next: verify_object_4073952c

VERIFICATION: Verify located object is blue box

tools: vision_system

failure modes: **verification_failure**

↳ next: pick_blue_box_f76bd4e3

ACTION: Pick blue box

tools: gripper

failure modes: **grasp_failure**

↳ next: move_to_workstation_2_446efc25

ACTION: Move to workstation 2

tools: mobile_base

failure modes: **timeout**

↳ next: place_blue_box_e81e2b8f

ACTION: Place blue box on workstation 2

tools: gripper

failure modes: **placement_failure**

↳ next: verify_placement_b5b00b04

VERIFICATION: Verify placement

tools: vision_system

failure modes: **placement_failure**

RECOVERY: Recovery: rescan shelf A for blue box

strategy: rescan → returns to locate_blue_box_930b5e05

RECOVERY: Recovery: retry grasp of blue box

strategy: retry_with_backoff → returns to pick_blue_box_f76bd4e3

RECOVERY: Recovery: retry navigation to workstation 2

strategy: retry → returns to move_to_workstation_2_446efc25

RECOVERY: Recovery: retry placement of blue box

strategy: retry → returns to place_blue_box_e81e2b8f

Simulate execution

INJECT FAILURE AT NODE (OPTIONAL)

Locate blue box at shelf A – **object_missing**

Run simulation

Execution trace **COMPLETED**

run: 4f1fdec7 steps: 18 total simulated latency: 2076 ms **recovered in 7 step(s)**

ENTER	Locate blue box at shelf A	Entering 'Locate blue box at shelf A'	
FAILURE	Locate blue box at shelf A	'Locate blue box at shelf A' failed with object_missing	object_missing
RECOVERY_TRIGGERED	Locate blue box at shelf A	Routing to recovery node 'rescan_790f23f1' for 'object_missing'.	object_missing
ENTER	Recovery: rescan shelf A f...	Entering 'Recovery: rescan shelf A for blue box'	
RETRY	Recovery: rescan shelf A f...	Recovery strategy 'rescan' executing	
ENTER	Locate blue box at shelf A	Entering 'Locate blue box at shelf A'	
SUCCESS	Locate blue box at shelf A	'Locate blue box at shelf A' succeeded	

